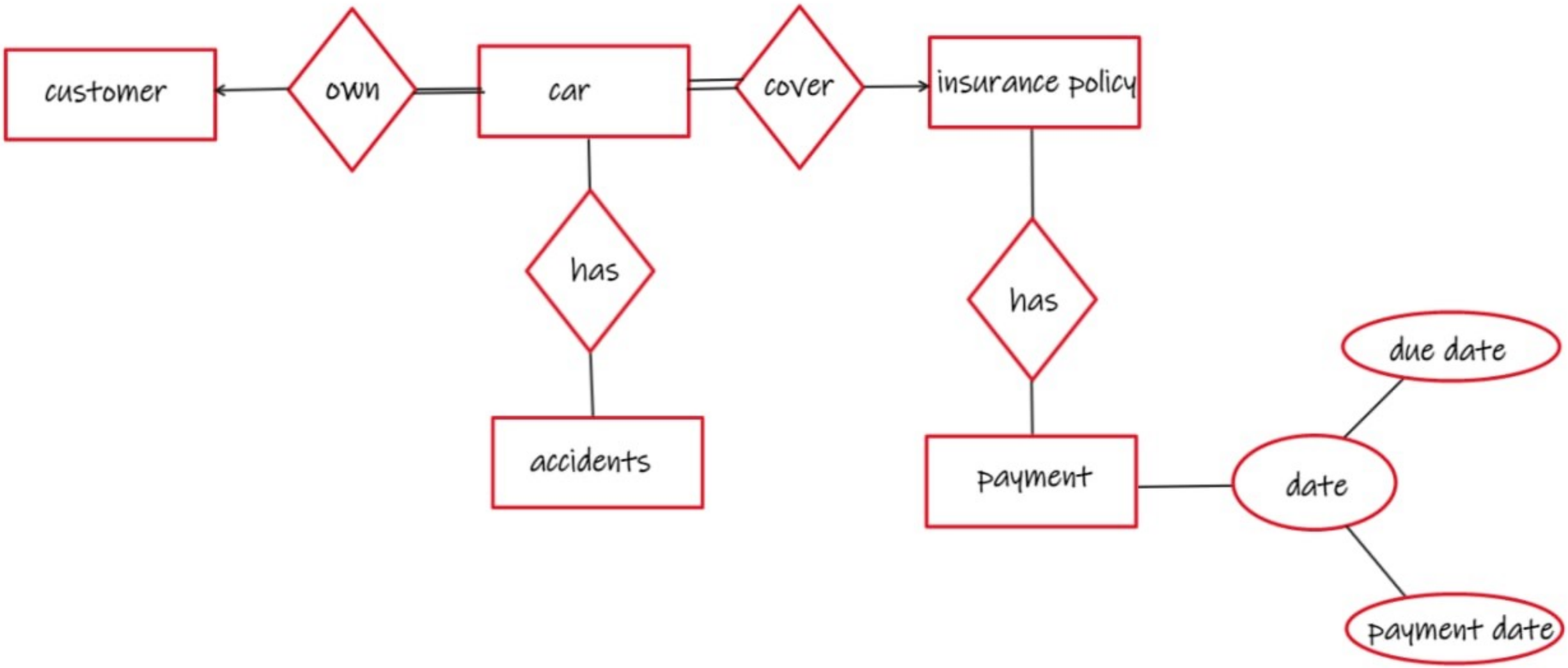
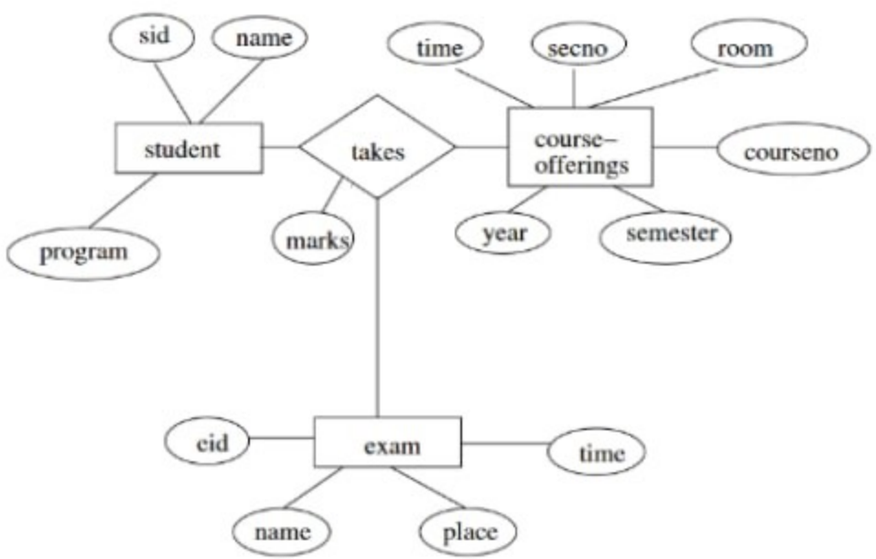
6.1 Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period of time, and has an associated due date, and the date when the payment was received.

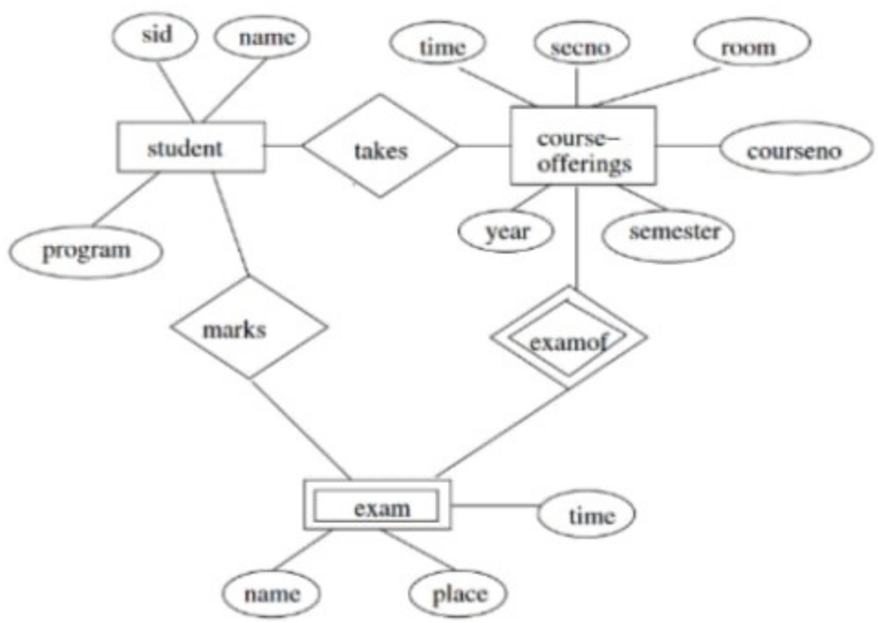


6.2 Consider a database that includes the entity sets student, course, and section
from the university schema and that additionally records the marks that students
receive in different exams of different sections.
a. Construct an E-R diagram that models exams as entities and uses a ternary relationship as part of the design.
b. Construct an alternative E-R diagram that uses only a binary relationship between student and section. Make sure that only one relationship exists between a particular student and section pair, yet you can represent the marks that a student gets in different exams.

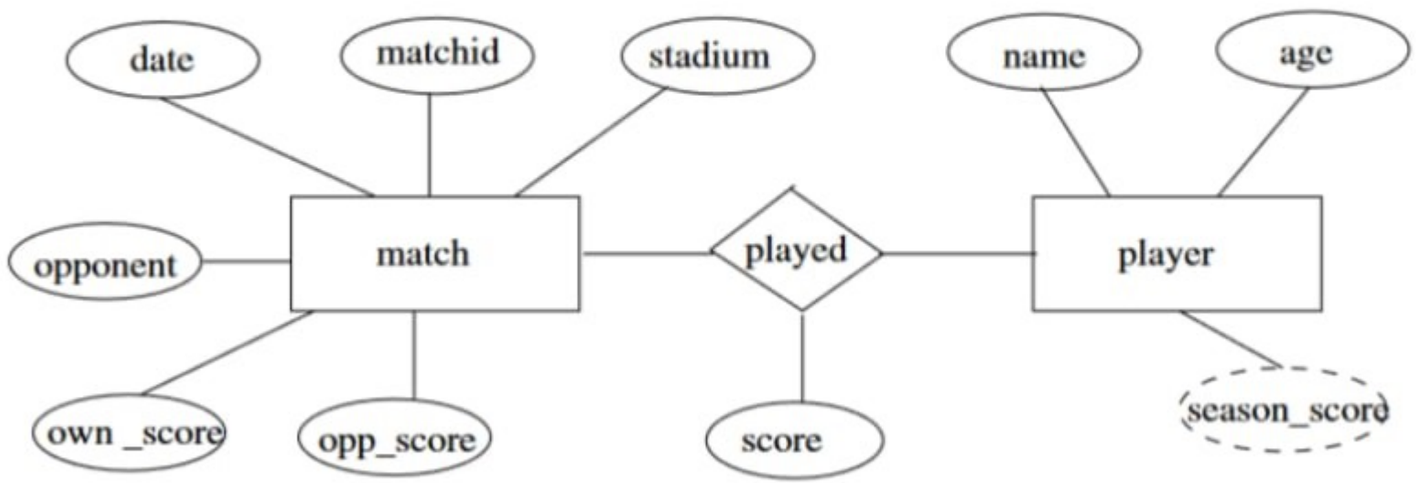
a)



b)



6.3 Design an E-R diagram for keeping track of the scoring statistics of your favorite sports team. You should store the matches played, the scores in each match, the players in each match, and individual player scoring statistics for each match. Summary statistics should be modeled as derived attributes with an explanation as to how they are computed



6.4 Consider an E-R diagram in which the same entity set appears several times, with its attributes repeated in more than one occurrence. Why is allowing this redundancy a bad practice that one should avoid?

By using one entity set many times we are missing relationships in the model ,for example employee could have many projects but also manage one department

6.5 An E-R diagram can be viewed as a graph. What do the following mean in terms of the structure of an enterprise schema?
a. The graph is disconnected.
b. The graph has a cycle.

a)A disconnected graph implies that there are pairs of entity sets that are unrelated to each other, hence maybe better off to design separate databases (each corresponding to a connected subgraph)

b)An acyclic graph means that there is a unique path between every pair of entity sets and, thus, a unique relationship between every pair of entity sets